THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Which of the following is a pure substance? 1

(a) sea water **(b)** distilled water **(c)** air **(d)** milk

2. The law of constant proportions was proposed by: 1

(a) Dalton **(b)** Joseph Proust **(c)** Lavoisier **(d)** Rutherford

3. The charge on an electron is: 1

(a) positive **(b)** negative **(c)** zero **(d)** variable

4. The distance travelled by a body is always: 1

(a) equal to velocity
(b) negative
(c) greater than or equal to the magnitude of displacement
(d) less than displacement

5. Inertia of a body depends on its: 1

(a) colour **(b)** speed only **(c)** volume **(d)** mass

6. The SI unit of work is the: 1

(a) newton **(b)** watt **(c)** pascal **(d)** joule

7. Sound cannot travel through: 1

(a) air **(b)** water **(c)** a vacuum **(d)** steel

8. Cell theory was proposed by: 1

(a) Darwin and Wallace **(b)** Hooke and Brown
(c) Watson and Crick **(d)** Schleiden and Schwann

9. Muscles attached to bones and used in running and writing are: 1

(a) cardiac muscles **(b)** striated (voluntary) muscles
(c) involuntary branched muscles **(d)** smooth muscles

10. Fish breathe through: 1

(a) gills **(b)** skin only **(c)** lungs **(d)** trachea

11. Sperms are produced in the: 1

(a) testes **(b)** ovaries **(c)** uterus **(d)** oviduct

12. The continuous movement of water between the oceans, atmosphere and land is the: 1

(a) rock cycle **(b)** water (hydrological) cycle
(c) nitrogen cycle **(d)** carbon cycle

13. One breath is about 0.5 litre and a person takes about 20000 breaths a day. The air breathed in a day is about: 1

(a) 100 litres **(b)** 1000 litres **(c)** 10000 litres **(d)** 100000 litres

14. Which of the following is a colloid? 1

(a) air **(b)** sugar in water **(c)** salt in water **(d)** fog

15. The atomicity of sulphur (S_8) is: 1

(a) 1 **(b)** 8 **(c)** 2 **(d)** 4

16. The mass number of an atom equals the number of: 1

(a) neutrons plus electrons **(b)** protons plus neutrons
(c) protons only **(d)** protons plus electrons

17. The area under a velocity-time graph gives: 1

(a) time **(b)** acceleration **(c)** displacement **(d)** speed

18. Action and reaction do not cancel because they: 1

(a) act at different times **(b)** are in the same direction
(c) act on different bodies **(d)** are unequal

19. The law of conservation of energy states that energy: 1

(a) can neither be created nor destroyed but only changed from one form to another
(b) can be destroyed
(c) is always lost
(d) can be created

20. The pitch of a sound depends on its: 1

(a) medium **(b)** amplitude **(c)** frequency **(d)** speed

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Name the four spheres of the Earth system and state one feature of each. 2

22. Why is ignoring air resistance not a mistake when studying a falling ball? 2

23. What is centrifugation? Give one use. 2

24. What is atomicity? Give the atomicity of H_2, O_3 and P_4. 2

25. Calculate the number of protons, neutrons and electrons in $^{23}_{11}Na$. 2

26. Define acceleration. Give its SI unit. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Differentiate between isotopes and isobars (three points). 3

28. From a velocity-time graph of a body moving at 6 m/s for 10 s find the distance covered. 3

29. A force of 12 N acts for 3 s on a body at rest of mass 4 kg. Find the velocity acquired. 3

30. Calculate the energy consumed and the cost if a 100 W bulb is used for 10 hours and the rate is 6 rupees per unit. 3

31. Explain the working of SONAR and calculate the depth of the sea if the echo of an ultrasound pulse is received after 4 s (speed in water = 1500 m/s). 3

OR

Explain why red blood cells and nerve cells have different shapes. 3

32. Describe the structure of a neuron with a labelled description. 3

OR

Describe the characteristics of Phylum Porifera and Cnidaria. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. A body of mass 5 kg is moving on a rough surface with an applied force of 30 N and friction of 10 N. Find the acceleration, the velocity after 4 s and the distance covered in 4 s starting from rest. 5

34. Explain the conservation of energy in a swinging pendulum. Describe the changes of KE and PE and the effect of air resistance. 5

35. Explain reverberation, its causes and effects in an auditorium, and the methods of controlling it, with the principle of curved sound boards. 5

36. Describe the two types of cell division and their significance. 5

OR

Describe the muscular tissue with the three types, structure and function of each in a table of five points. 5

37. Compare monocots and dicots in a table of five points with examples. 5

OR

Compare asexual and sexual reproduction in five points with examples. 5

38. Describe the nitrogen cycle with the processes: fixation, absorption, assimilation, decay, and denitrification. 5

OR

Discuss the role of models in science with three examples from chemistry or physics, and explain how models change. 5

END OF QUESTION PAPER — Total: 80 Marks

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